ros2log new CLI and rcl_logging_implementation for Lyrical Luth



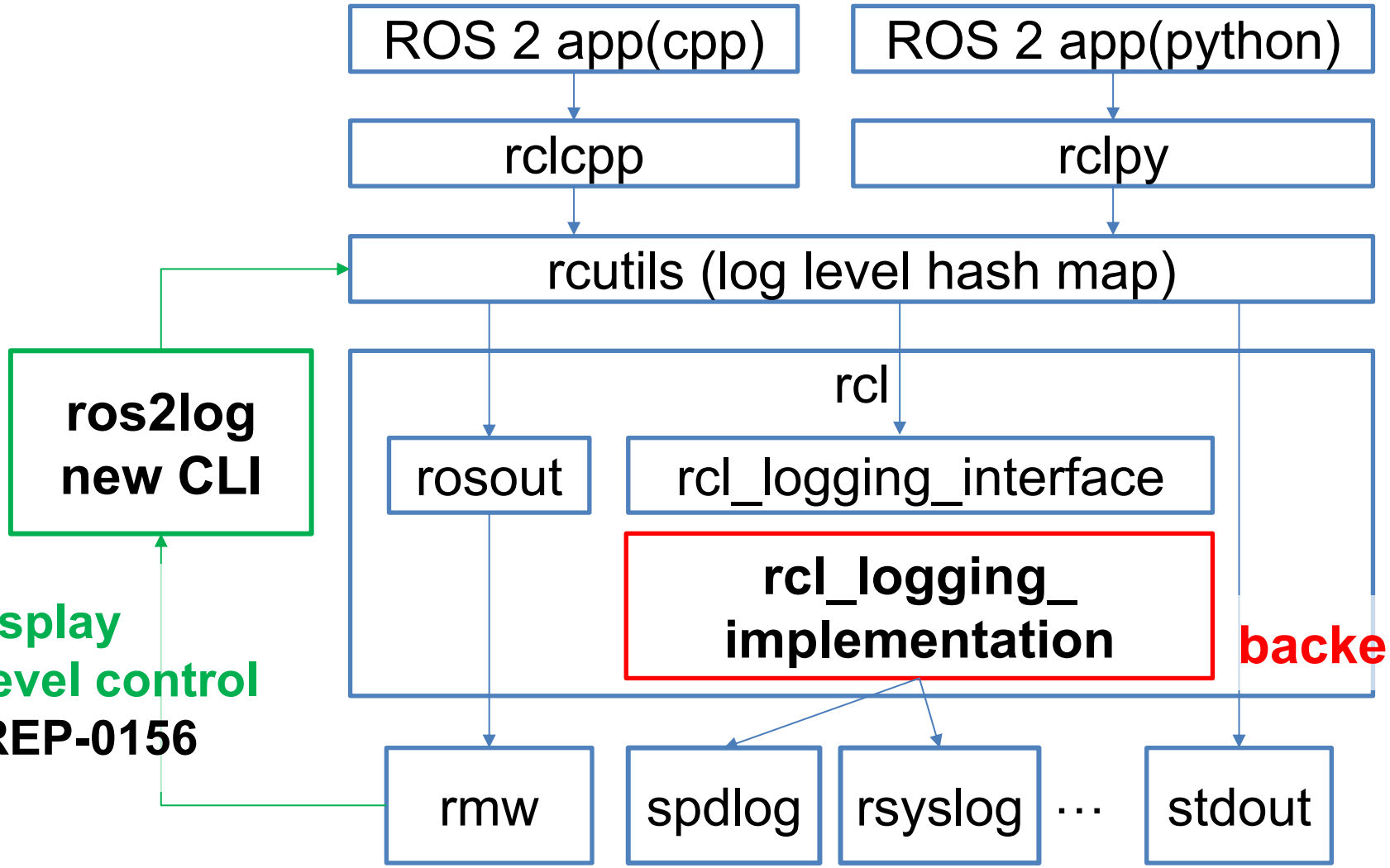




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TriOrb.Inc

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Keio University

Overview

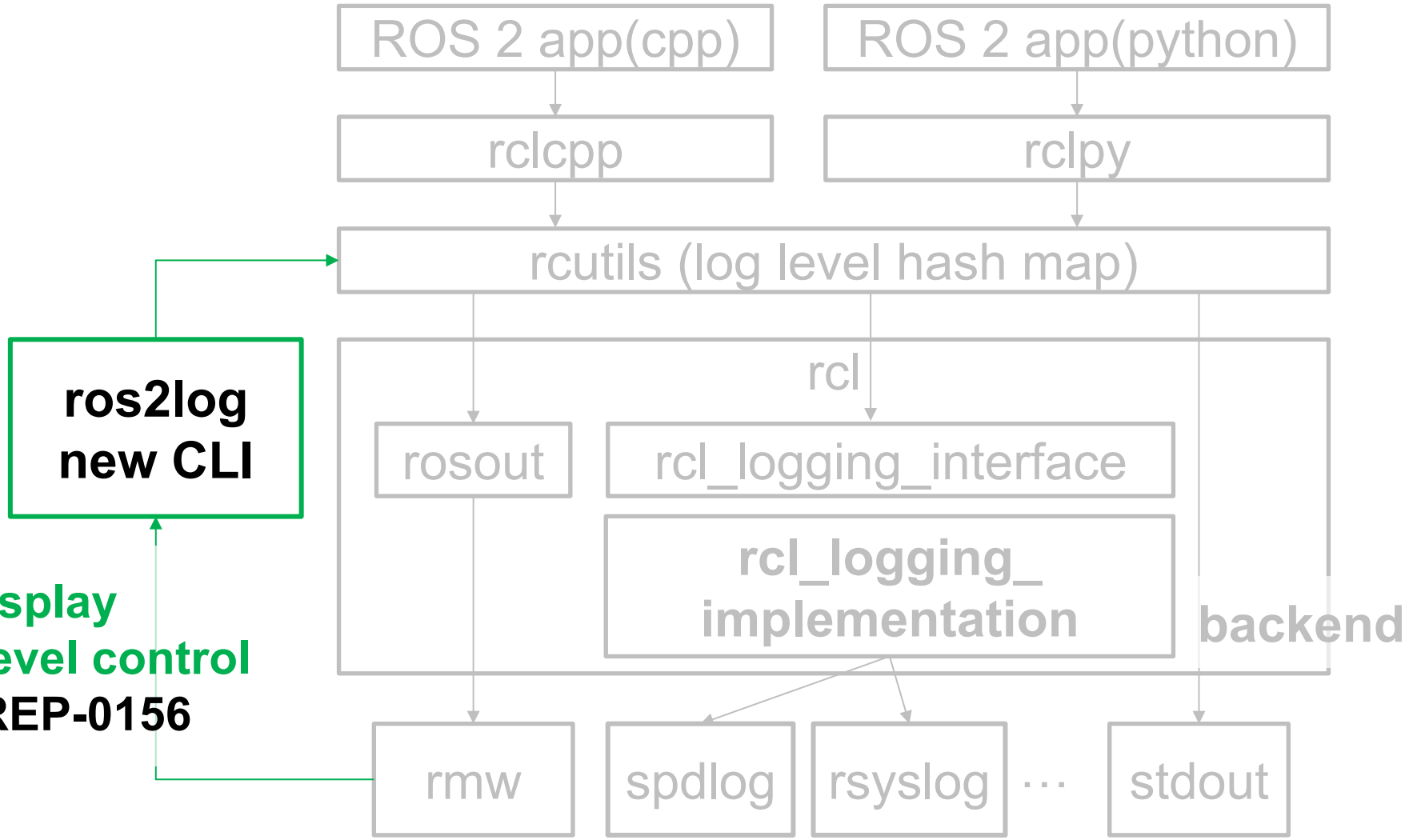


- **Filtered log display**
- **Runtime log level control**

➢ Proposed as **REP-0156**

- Runtime dynamic loading of logging backends **without rebuilding RCL or application code**

Overview



frontend

ros2log
new CLI

- **Filtered log display**
- **Runtime log level control**

➢ Proposed as **REP-0156**

- Runtime dynamic loading of logging backends **without rebuilding RCL or application code**

Background


Windows PowerShell
decwest@ytpc2022e: ~/decwest_v
decwest@ytpc2022e: ~/decwest_
[rviz2-5] [ERROR] [1785313777.068921926] [rviz2]: Could not load pixmap package://rviz_common/icons/classes/Displays.svg
[rviz2-5] [ERROR] [1785313777.072177175] [rviz2]: Could not load pixmap package://rviz_common/icons/classes/Views.svg
[planner_server-9] [INFO] [1785313777.076068239] [planner_server]: Created global planner plugin GridBased of type nav2_navfn_planner::NavfnPl
er
[planner_server-9] [INFO] [1785313777.076119854] [planner_server]: Configuring plugin GridBased of t
[planner_server-9] [INFO] [1785313777.076433012] [planner_server]: Planner Server has GridBased pla
[lifecycle_manager-18] [INFO] [1785313777.100358794] [lifecycle_manager_navigation]: Configuring rou
[route_server-10] [INFO] [1785313777.100635171] [route_server]: Configuring
[route_server-10] [INFO] [1785313777.161446025] [route_server]: Created GraphFileLoader GeoJsonGraph
leLoader
[route_server-10] [INFO] [1785313777.161487594] [route_server]: Configuring geojson graph file
[route_server-10] [INFO] [1785313777.161492702] [route_server]: No graph file
[route_server-10] [INFO] [1785313777.164071237] [route_server]: Created ed
er
[route_server-10] [INFO] [1785313777.164117841] [route_server]: Configuri
[route_server-10] [INFO] [1785313777.164207863] [route_server]: Cre
[route_server-10] [INFO] [1785313777.164215877] [route_server]: Co
[rviz2-5] [WARN] [1785313777.165289470] [rcl.logging_rosout]: Publi
es with the same name then all logs for the logger named 'rviz2' wil
estructed it will unregister the publisher, preventing any further
[route_server-10] [INFO] [1785313777.166954630] [route_server]: Crea
[route_server-10] [INFO] [1785313777.166995552] [route_server]: Configuri
[route_server-10] [INFO] [1785313777.168110294] [route_server]: Created route op
[route_server-10] [INFO] [1785313777.168140367] [route_server]: Configuring Rerouting ser
[route_server-10] [INFO] [1785313777.170329809] [route_server]: Created route operation Co
[route_server-10] [INFO] [1785313777.170427543] [route_server]: Using costmap topic: loca
lobal_costmap/costmap_raw for CollisionMonitor.
[lifecycle_manager-18] [INFO] [1785313777.174891237] [lifecycle_manager_navigation]: Configuring b
[behavior_server-11] [INFO] [1785313777.175176256] [behavior_server]: Configuring
[behavior_server-11] [INFO] [1785313777.180786972] [behavior_server]: Creating behavior plugin spi
[rviz2-5] [INFO] [1785313777.180857313] [rviz2]: Trying to load plugins...
[behavior_server-11] [INFO] [1785313777.182088904] [behavior_server]: Creating behavior plugin bac
[rviz2-5] [WARN] [1785313777.183139946] [rcl.logging_rosout]: Publisher already registered for noc
es with the same name then all logs for the logger named 'rviz2' will go out over the existing pub
estructed it will unregister the publisher, preventing any further logs for that name from being p
[behavior_server-11] [INFO] [1785313777.183618223] [behavior_server]: Creating behavior plugin dri
ading
[behavior_server-11] [INFO] [1785313777.184762414] [behavior_server]: Creating behavior plugin ass
leop
PathB
• All logs shown at once, hard to read…
• Mixed log levels…

Background

- Changing log levels at runtime

```
ros2 service call ¥
  <node_name>/set_logger_levels ¥
  rcl_interfaces/srv/SetLoggerLevels ¥
  "{levels: [{name: <logger>, level: <level>}]}"
```

  - ☹ typing this by hand every time is tedious…

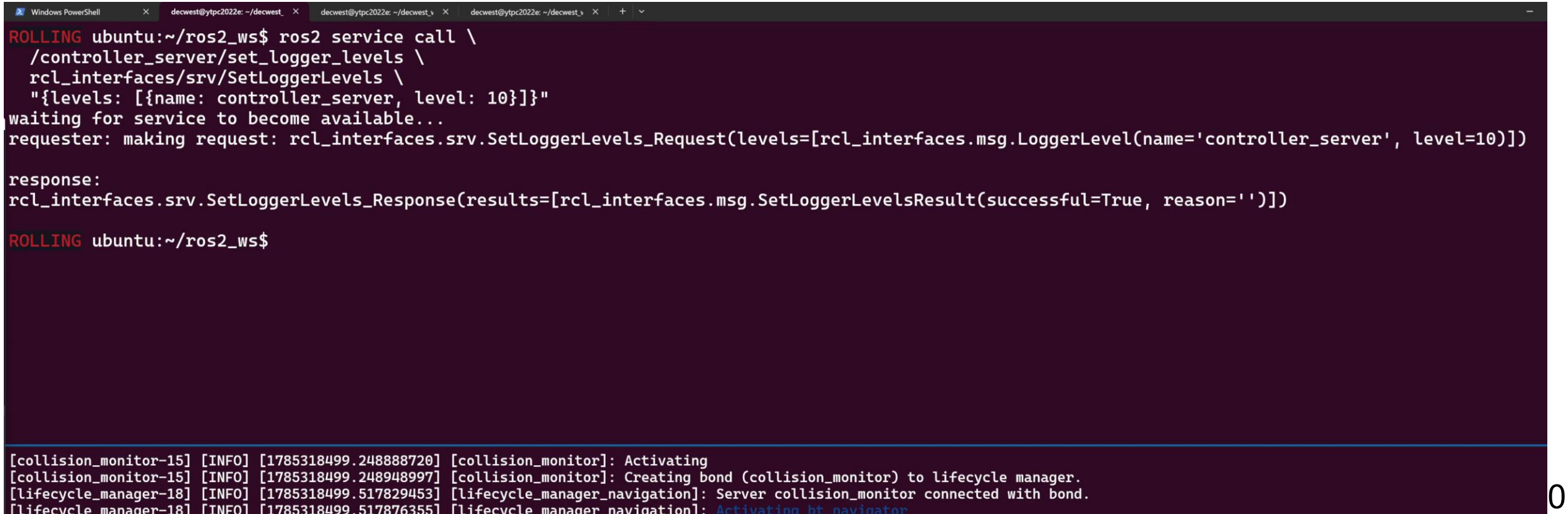
Note: runtime log level change applies only to nodes created with **`enable_logger_service=true`**

- **Python**

```python
node = Node('NodeWithLoggerService',
  enable_logger_service=True)
```

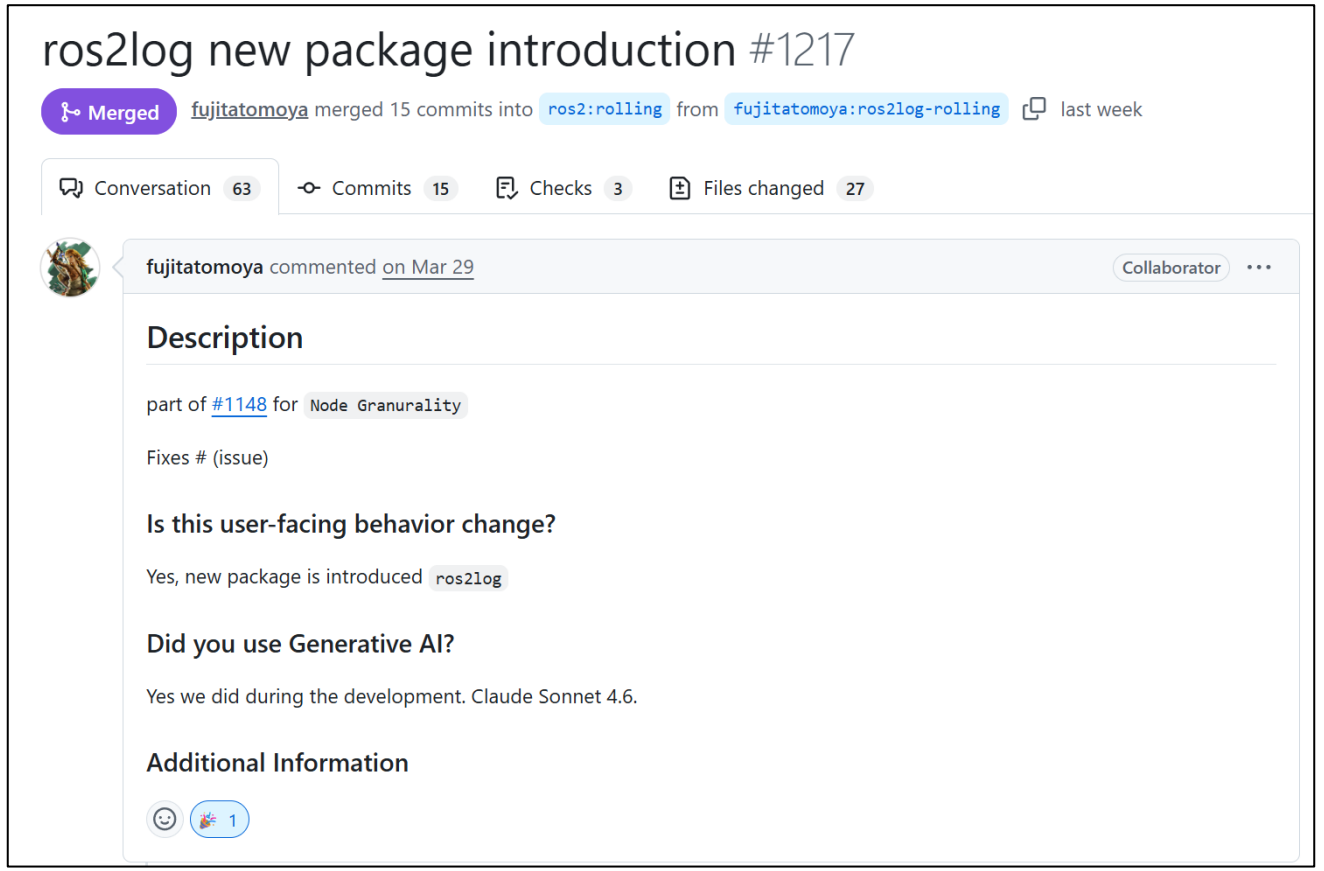
- **C++**

```cpp
auto node =
std::make_shared<rclcpp::Node>
  ("NodeWithLoggerService",
  rclcpp::NodeOptions()
    .enable_logger_service(true))
```



Our Contributions

- ✓ Developed the CLI **ros2log** to unify ROS 2 log control
- ✓ Proposed as **REP-0156** in the official standardization process



ros2log new package introduction #1217

Merged fujitatomoya merged 15 commits into ros2:rolling from fujitatomoya:ros2log-rolling last week

Conversation 63 Commits 15 Checks 3 Files changed 27

fujitatomoya commented on Mar 29 Collaborator

Description

part of #1148 for `Node Granurality`

Fixes # (issue)

Is this user-facing behavior change?

Yes, new package is introduced `ros2log`

Did you use Generative AI?

Yes we did during the development. Claude Sonnet 4.6.

Additional Information

1

Available from ROS 2 Lyrical onward 🚀

Usage Example

- ros2 log watch --regex controller

```
Windows PowerShell   decwest@ytpc2022e: ~/decwest_   decwest@ytpc2022e: ~/decwest_v
[1785313115.351035] [INFO] [controller_server] : Created progress_checker : progress_checker of type nav2_controller::SimpleProgressChecker
[1785313115.352000] [INFO] [controller_server] : Created goal checker : general_goal_checker of type nav2_controller::SimpleGoalChecker
[1785313115.353439] [INFO] [controller_server] : Created path handler : PathHandler of type nav2_controller::FeasiblePathHandler
[1785313115.354748] [INFO] [controller_server] : Created controller : PP of type nav2_regulated_pure_pursuit_controller::RegulatedPurePursuitControl
[1785313115.361511] [INFO] [controller_server] : Created controller : APP of type nav2_regulated_pure_pursuit_controller::RegulatedPurePursuitContro
r
[1785313115.365091] [INFO] [controller_server] : Created controller : RPP of type nav2_regulated_pure_pursuit_controller::RegulatedPurePursuitContro
r
[1785313115.368800] [INFO] [controller_server] : Created controller : DWPP of type nav2_regulated_pure_pursuit_controller::RegulatedPurePursuitContr
er
[1785313115.372521] [INFO] [controller_server] : Controller Server has PP APP RPP DWPP  controllers available.
[1785313115.286647] [INFO] [lifecycle_manager_navigation] : Configuring controller_server
[1785313116.028291] [INFO] [lifecycle_manager_navigation] : Activating controller_server
[1785313117.629101] [INFO] [controller_server] : Activating controller: DWPP of type regulated_pure_pursuit_controller::RegulatedPurePursuitControll
[1785313117.629338] [INFO] [controller_server] : Activating controller: RPP of type regulated_pure_pursuit_controller::RegulatedPurePursuitControlle
[1785313117.629394] [INFO] [controller_server] : Activating controller: APP of type regulated_pure_pursuit_controller::RegulatedPurePursuitControlle
[1785313117.629404] [INFO] [controller_server] : Activating controller: PP of type regulated_pure_pursuit_controller::RegulatedPurePursuitController
[1785313117.629828] [INFO] [controller_server] : Creating bond (controller_server) to lifecycle manager.
[1785313117.893327] [INFO] [lifecycle_manager_navigation] : Server controller_server connected with bond.
```

Filter logs!

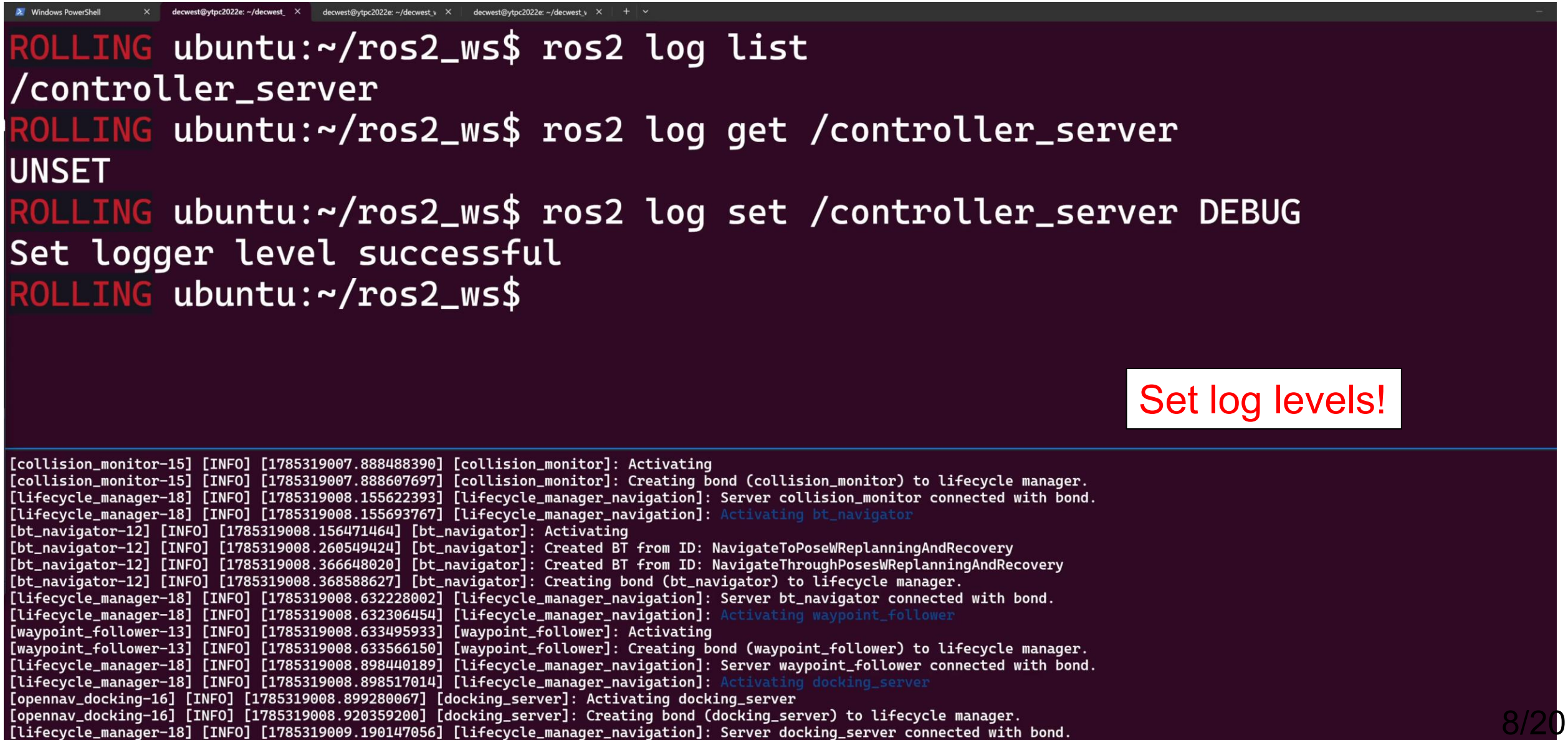
```
[collision_monitor-15] [INFO] [1785313119.281568094] [collision_monitor]: Activating
[collision_monitor-15] [INFO] [1785313119.281713461] [collision_monitor]: Creating bond (collision_monitor) to lifecycle manager.
[lifecycle_manager-18] [INFO] [1785313119.548456535] [lifecycle_manager_navigation]: Server collision_monitor connected with bond.
[lifecycle_manager-18] [INFO] [1785313119.548691189] [lifecycle_manager_navigation]: Activating bt_navigator
[bt_navigator-12] [INFO] [1785313119.549451047] [bt_navigator]: Activating
[bt_navigator-12] [INFO] [1785313119.647398647] [bt_navigator]: Created BT from ID: NavigateToPoseWReplanningAndRecovery
[bt_navigator-12] [INFO] [1785313119.733050640] [bt_navigator]: Created BT from ID: NavigateThroughPosesWReplanningAndRecovery
[bt_navigator-12] [INFO] [1785313119.734624340] [bt_navigator]: Creating bond (bt_navigator) to lifecycle manager.
[lifecycle_manager-18] [INFO] [1785313119.998101004] [lifecycle_manager_navigation]: Server bt_navigator connected with bond.
[lifecycle_manager-18] [INFO] [1785313119.998411633] [lifecycle_manager_navigation]: Activating waypoint_follower
[waypoint_follower-13] [INFO] [1785313119.998810624] [waypoint_follower]: Activating
[waypoint_follower-13] [INFO] [1785313119.998884724] [waypoint_follower]: Creating bond (waypoint_follower) to lifecycle manager.
[lifecycle_manager-18] [INFO] [1785313120.259302512] [lifecycle_manager_navigation]: Server waypoint_follower connected with bond.
[lifecycle_manager-18] [INFO] [1785313120.259466820] [lifecycle_manager_navigation]: Activating docking_server
[opennav_docking-16] [INFO] [1785313120.260962485] [docking_server]: Activating docking_server
[opennav_docking-16] [INFO] [1785313120.283574406] [docking_server]: Creating bond (docking_server) to lifecycle manager.
[lifecycle_manager-18] [INFO] [1785313120.550152658] [lifecycle_manager_navigation]: Server docking_server connected with bond.
```

Usage Example

Old command →

```
ros2 service call ¥
  /controller_server/set_logger_levels ¥
  rcl_interfaces/srv/SetLoggerLevels ¥
  "{levels: [{name: controller_server, level: 10}]}"
```

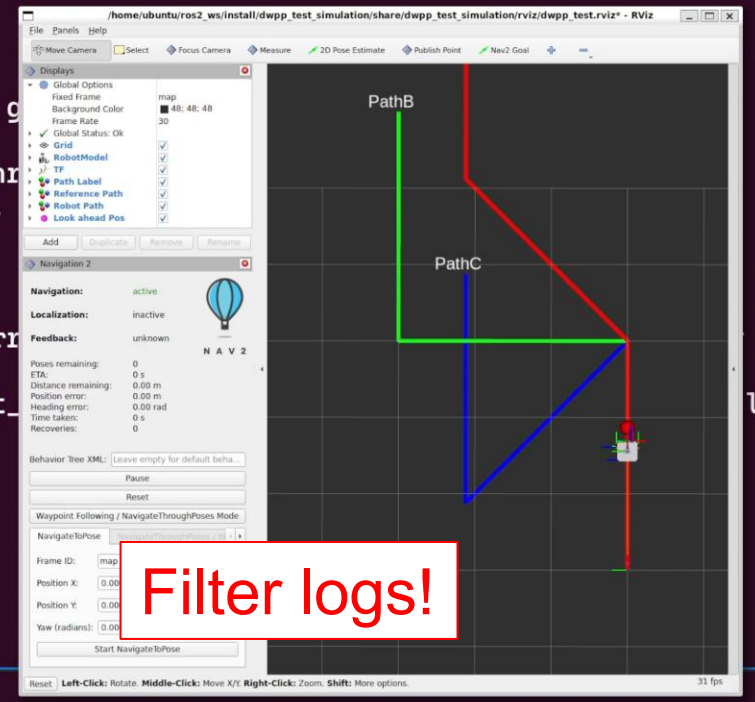
- ros2 log set /controller_server DEBUG



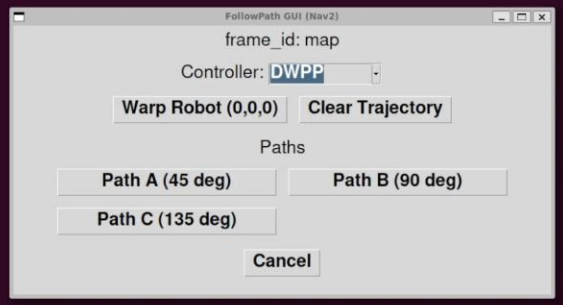
Usage Example

- ros2 log watch --logger controller_server --level DEBUG

```
ROLLING ubuntu:~/ros2_ws$ ros2 log set /controller_server DEBUG
Set logger level successful
ROLLING ubuntu:~/ros2_ws$ ros2 log watch --logger controller_server --level DEBUG
[1785319360.279507] [DEBUG] [controller_server] : [follow_path] [ActionServer] Received request for g
[1785319360.280104] [DEBUG] [controller_server] : [follow_path] [ActionServer] Receiving a new goal
[1785319360.280164] [DEBUG] [controller_server] : [follow_path] [ActionServer] Executing goal asynchr
[1785319360.280452] [DEBUG] [controller_server] : [follow_path] [ActionServer] Executing the goal...
[1785319360.280630] [INFO] [controller_server] : Received a goal, begin computing control effort.
[1785319360.280696] [DEBUG] [controller_server] : Selected controller: DWPP.
[1785319360.280750] [DEBUG] [controller_server] : Selected goal checker: general_goal_checker.
[1785319360.280800] [WARN] [controller_server] : No progress checker was specified in parameter 'curr
 loaded progress_checker . This warning will appear once.
[1785319360.280855] [WARN] [controller_server] : No path handler was specified in parameter 'current_
PathHandler . This warning will appear once.
[1785319360.280890] [DEBUG] [controller_server] : Providing path to the controller DWPP
[1785319360.281039] [DEBUG] [controller_server] : Path end point is (3.88, 2.12)
[1785319360.282268] [DEBUG] [controller_server] : Publishing velocity at time 34.02
[1785319360.315989] [DEBUG] [controller_server] : Publishing velocity at time 34.05
[1785319360.349490] [DEBUG] [controller_server] : Publishing velocity at time 34.08
[1785319360.382858] [DEBUG] [controller_server] : Publishing velocity at time 34.10
```




```
[controller_server-7] [DEBUG] [1785319368.181899144] [controller_server]: Publishing velocity at time 40.14
[controller_server-7] [DEBUG] [1785319368.216202968] [controller_server]: Publishing velocity at time 40.16
[controller_server-7] [DEBUG] [1785319368.248617209] [controller_server]: Publishing velocity at time 40.19
[controller_server-7] [DEBUG] [1785319368.282925188] [controller_server]: Publishing velocity at time 40.22
[controller_server-7] [DEBUG] [1785319368.315584933] [controller_server]: Publishing velocity at time 40.24
[controller_server-7] [DEBUG] [1785319368.349888230] [controller_server]: Publishing velocity at time 40.27
[controller_server-7] [DEBUG] [1785319368.382542560] [controller_server]: Publishing velocity at time 40.30
[controller_server-7] [DEBUG] [1785319368.415837632] [controller_server]: Publishing velocity at time 40.32
[controller_server-7] [DEBUG] [1785319368.448806086] [controller_server]: Publishing velocity at time 40.35
[controller_server-7] [DEBUG] [1785319368.482880590] [controller_server]: Publishing velocity at time 40.38
[controller_server-7] [DEBUG] [1785319368.515378701] [controller_server]: Publishing velocity at time 40.40
[controller_server-7] [DEBUG] [1785319368.548859955] [controller_server]: Publishing velocity at time 40.43
[controller_server-7] [DEBUG] [1785319368.582343566] [controller_server]: Publishing velocity at time 40.45
[controller_server-7] [DEBUG] [1785319368.615780613] [controller_server]: Publishing velocity at time 40.48
[controller_server-7] [DEBUG] [1785319368.649105354] [controller_server]: Publishing velocity at time 40.51
[controller_server-7] [DEBUG] [1785319368.682721801] [controller_server]: Publishing velocity at time 40.53
[controller_server-7] [DEBUG] [1785319368.716499827] [controller_server]: Publishing velocity at time 40.56
```



ros2log core features defined by REP-0156

- Command
  - **ros2 log <subcommand>**
- Subcommands
  - **watch [options]**
    - **Subscribes to /rosout topic and prints logs filtered by** [options]
      - --level: log level
      - --regex: regular expression
      - --logger: logger name (node name in current impl.)

    Log filter
  - **levels**
    - Prints available log levels (UNSET, DEBUG, INFO, WARN, ERROR, FATAL)
  - **list**
    - Lists nodes that support runtime log level change
  - **get <node>**
    - Gets the current log level of <node>
  - **set <node> <level>**
    - Sets the log level of <node> to <level>

    Log-Level control

REP-0156
Documentation

Implementation Details

- Where ros2log is developed
  - **ros2cli**: https://github.com/ros2/ros2cli
  - Implementation language: **Python**
    - ros2cli is a Python CLI framework
- Key points of implementation
  - watch
    - ➢ **Subscribes to /rosout topic**
    - ➢ **Content Filtered Topics** are used
  - list, get, set
    - ➢ **/node/get_logger_levels**, **/node/set_logger_levels services** are used (available when a node is created with enable_logger_service=True)
    - list: lists nodes that have get, set services
    - get: wrapper of the get service
    - set: wrapper of the set service

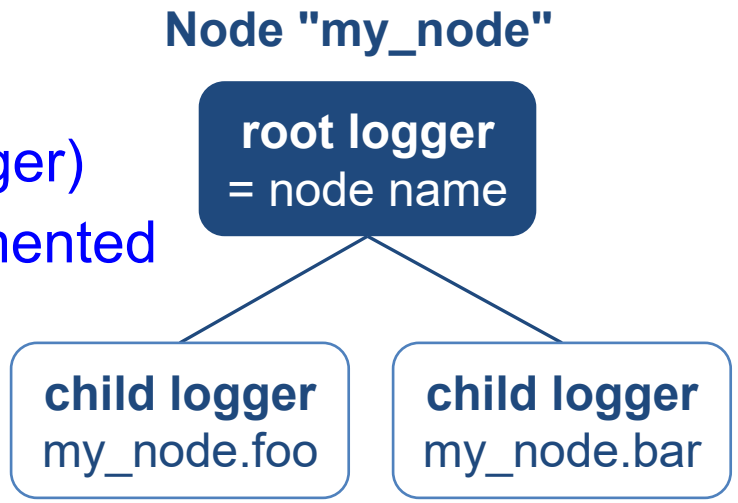
What are Content Filtered Topics?

- **Lets a subscriber receive only messages matching content conditions** (uses DDS ContentFilteredTopic via the RMW layer)
- Reduces reception overhead and traffic

Limitation & Future Work

- About log filtering
  - ➢ **Logs not in the /rosout topic cannot be shown**
    - Launch system's own logs, OS/device syslog, past logs, etc.…
  - ➞ **`ros2 log watch` targets only runtime logs from /rosout**
    - Log storage format/destination/on-off may vary per run with logging backend and user settings

- About log level control
  - ➢ Hard to control loggers besides the root logger (i.e. child logger)
  - ➢ Getting logger info via a `describe` subcommand: not implemented
  - – Reason: ROS 2 has no API to enumerate **logger names**
  - ➞ **Add an API to enumerate logger names, then support `describe` subcommand and child logger control**



Child logger

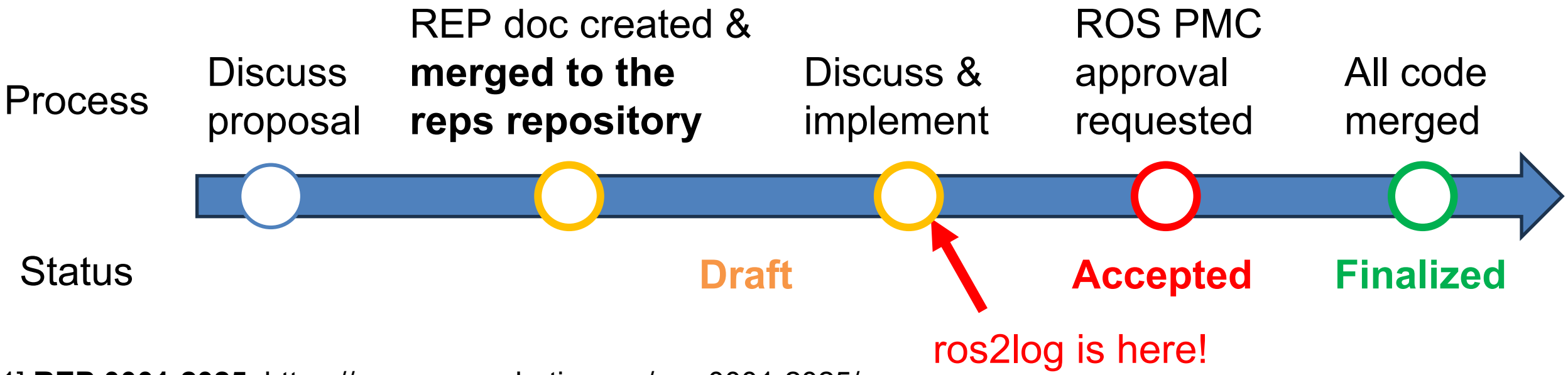
What is “REP”?

- **REP (Robotics Enhancement Proposal)**
  - A document explaining the what, how, and why of enhancements to **OSRA-governed projects**
  - Targets large changes spanning multiple PRs and repos
  - **Renamed in Oct. 2025** (was: ROS Enhancement Proposal)

- ROS
- Gazebo
- Open-RMF
- ROS Infrastructure
- ros_control

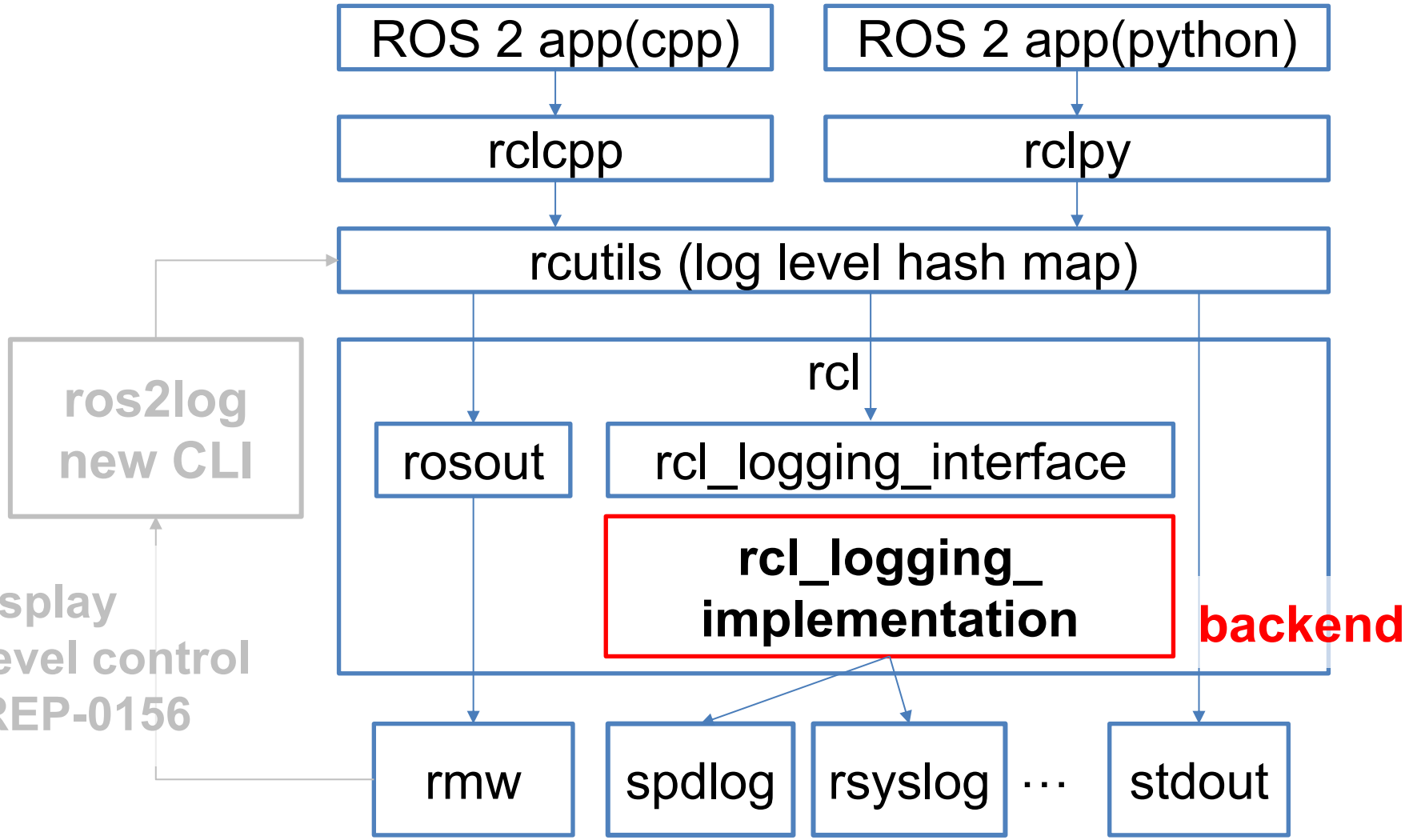
OSRA-governed OSS projects

- **Standardization process based on REP** [1]



[1] **REP 0001:2025**, https://reps.openrobotics.org/rep-0001-2025/

Overview



frontend

ros2log
new CLI

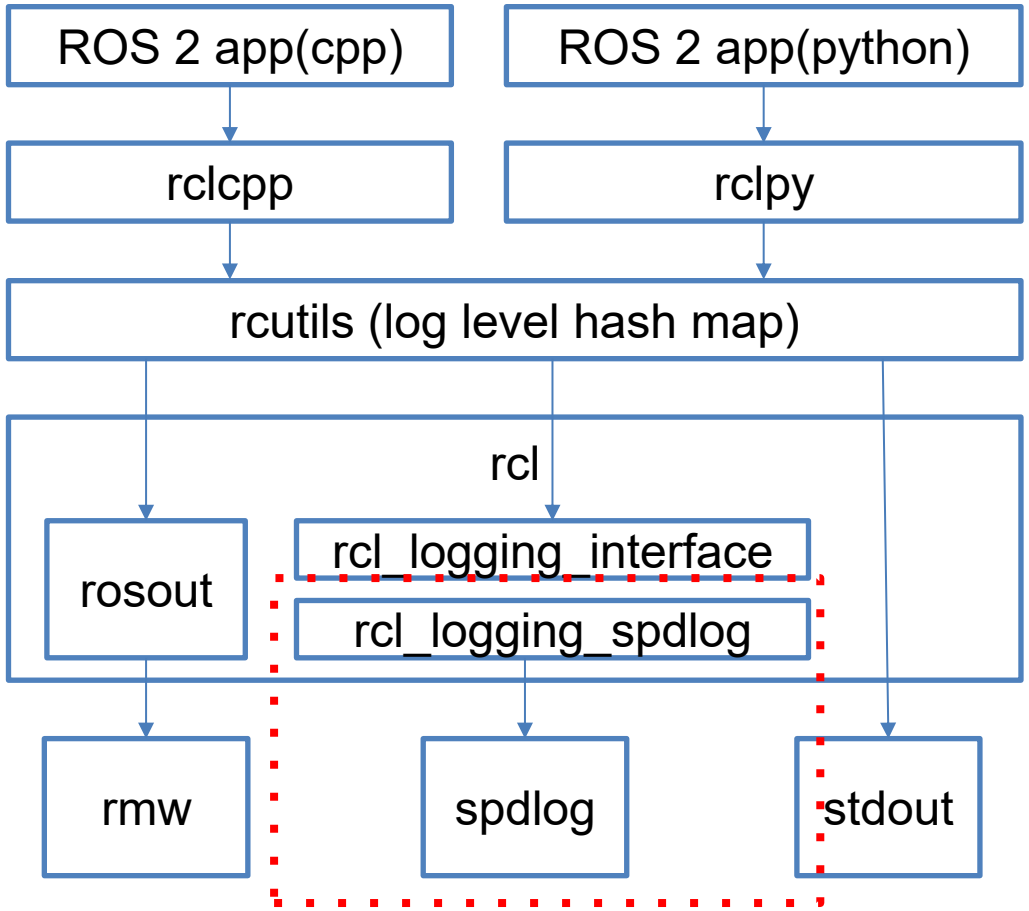
- Filtered log display
- Runtime log level control

➢ Proposed as **REP-0156**

- Runtime dynamic loading of logging backends **without rebuilding RCL or application code**

ROS 2 Logging Subsystem

- ROS 2 provides a flexible and configurable logging subsystem.
- Supports multiple logging backends.
  - **stdout**: Logs are printed to the standard output (default ON to console)
  - **rosout**: Logs are published to the /rosout topic (default ON)
  - **external**: Default spdlog, or your own implementation.
- A logger will only process log messages with severity at or higher than a specified level chosen for the logger.
- Logger names represent a hierarchy.

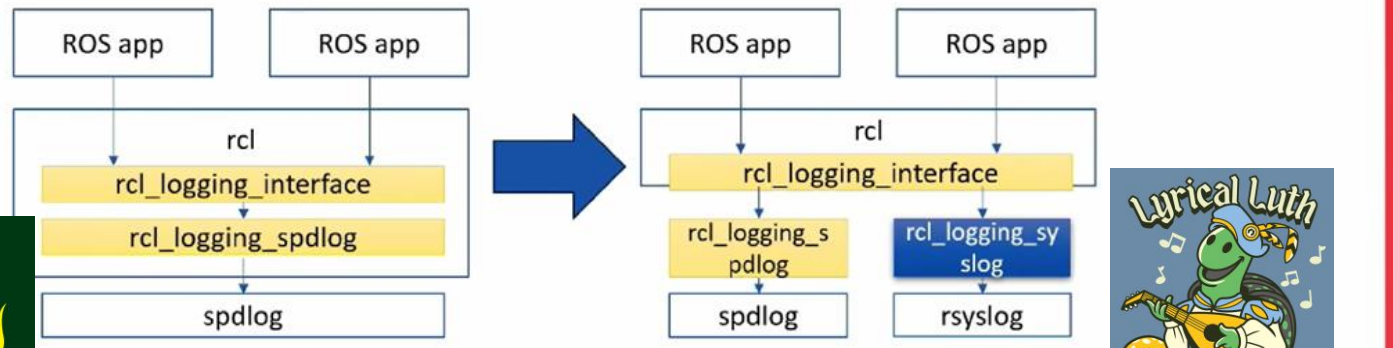


What was the pain?

rcl_logging_interface implementation must be build and integrated with rcl binary. That says, we cannot change the rcl logging implementation at runtime without rebuilding the source code. This needed to be fixed as [Feature Request: Select the logger without rebuilding]

Future Plan

- Package is not available yet, because...
- rcl_logging_interface implementation must be build and integrated with rcl binary. That says, we cannot change the rcl logging implementation at runtime without rebuilding the source code. This needs to be fixed Feature Request: Select the logger without rebuilding





rcl_logging_implementation

is a package that enables runtime dynamic loading of logging backends in ROS 2, similar to how rmw_implementation works for middleware selection. This abstraction layer allows users to switch between different logging implementations (such as rcl_logging_spdlog and rcl_logging_noop) without rebuilding RCL or application code.

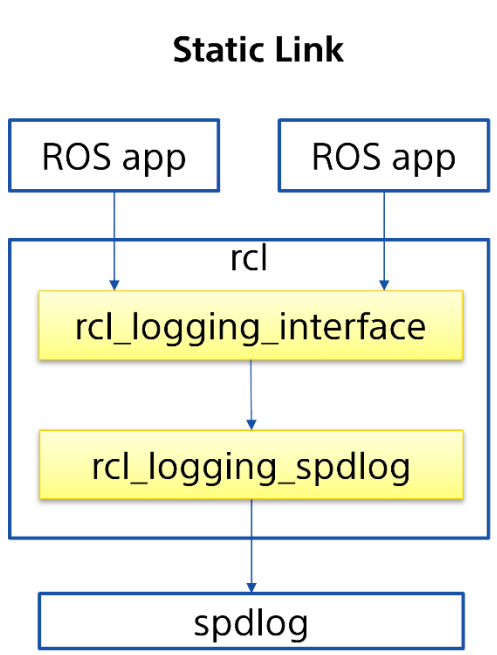
See more details for [About-Logging#rcl-logging-implementation]

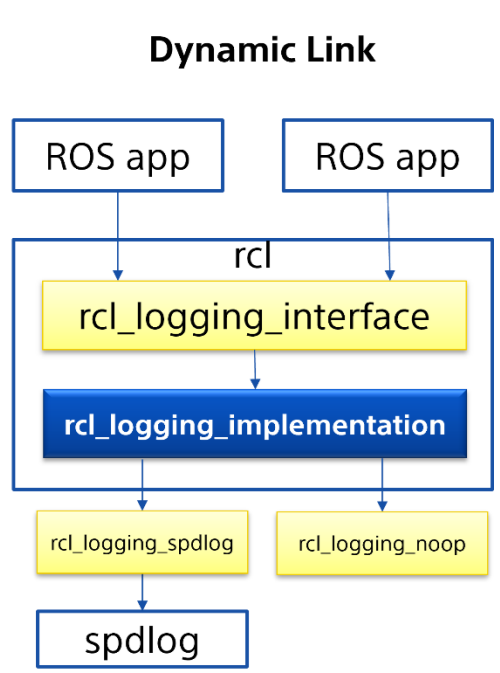
- **RCL_LOGGING_IMPLEMENTATION**
  - Dynamic Loading (default, runtime)
    - Switch logging implementation

      e.g) rcl_logging_noop
  - Static Link (Build time)
    - Bypass abstraction, No dlopen overhead
    - Bind rcl to specific logging implementation

```
# Use spdlog for logging (default behavior)
export RCL_LOGGING_IMPLEMENTATION=rcl_logging_spdlog
ros2 run demo_nodes_cpp talker

# Use no-op logging implementation that discards all log messages to the logging backend
export RCL_LOGGING_IMPLEMENTATION=rcl_logging_noop
ros2 run demo_nodes_cpp talker
```

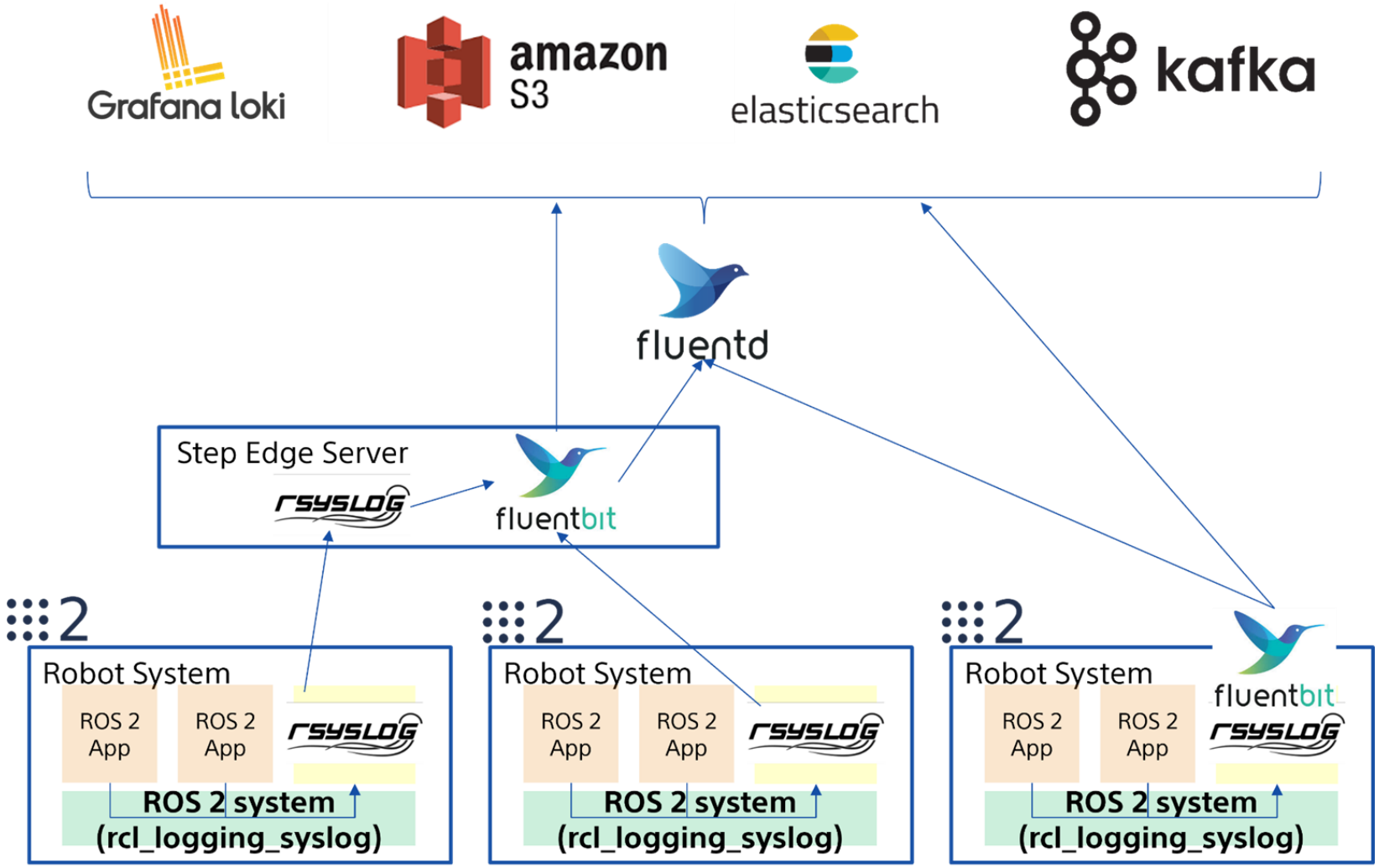




rcl_logging_syslog

- Alternate implementation for rcl_logging_interface
- ROS 2 rcl logging implementation built on top of syslog(3)
- Connects with rsyslog and FluentBit etc...

rcl_logging_syslog



How to use

We do not have to rebuild anymore !!!

```
apt install ros-lyrical-rcl-logging-syslog
export RCL_LOGGING_IMPLEMENTATION=rcl_logging_syslog
```

- If you use docker, either binding -v /dev/log:/dev/log or enable rsyslogd in the container.

- See more details for Installation Tutorial



Summary

- **ros2log**
  - Developed new CLI **ros2log** to unify ROS 2 log control
    - **REP-0156** under proposal (Status: Draft)
  - Future Work
    - Add an API to enumerate logger names
    - Support the `describe` subcommand and child logger control

    → Move REP-0156 to Finalized

- **ROS 2 logging subsystem**
  - rcl_logging_implementation has been completed
  - maybe journald support?
  - ROS 2 generic logging configuration translated into backend configuration

REP-0156
Documentation

rcl_logging_syslog